Environmental Innovation and Societal Transitions 3 (2012) 50–66

Contents lists available at SciVerse ScienceDirect

Environmental Innovation and Societal Transitions

journal homepage: www.elsevier.com/locate/eist

How do we govern sustainable innovations? Mapping patterns of governance for biofuels and hybrid-electric vehicle technologies

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ARTICLE INFO

Article history:
Received 19 January 2012
Received in revised form 22 April 2012
Accepted 22 April 2012
Available online 18 May 2012

Keywords:
Biofuels
Hybrid-electric vehicles
Innovation
Policy
Sustainability
Sweden
Transport

ABSTRACT

This paper examines patterns of governance aimed at sustainable technological innovation in the transport sector. It makes an overall assessment of governance emerging in the fields of biofuel and hybrid-electric vehicle (HEV) technologies, and makes a classification of its characteristics. It examines the role of different actors and levels of governance as well as preferred mechanisms and targets of governance. The assessment reveals that there are rather differential patterns of governance influencing the two fields. For instance, international-level and market-based governance are much more prevalent in biofuels, whereas industry-led and cognitive governance play comparatively stronger roles in HEV. These patterns can be understood in light of both the different institutional and actor characteristics of the two technologies, and their positions in relation to socio-technical regimes.

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1. Introduction

The quest for sustainable transport has emerged as a principal governance challenge. Transport is one of the sectors where environmental and resource pressures keep mounting over and above

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http://dx.doi.org/10.1016/j.eist.2012.04.002

already unsustainable levels. Contributing to this is continued growth in freight and personal transport, reliance on private automobiles, and continued use of fossil fuels with conventional internal combustion engine technologies. At the same time, much hope has been placed on the development and rapid uptake of new vehicle technologies such as the hydrogen-based fuel cell, biofuels and hybrid-electric vehicles (HEV). However, up until recently these new technologies have had difficulties to compete on the market, and further innovations are needed. Given the urgency to resolve unsustainable trends in the sector, today both industry and policy makers agree that innovation in these technologies requires more active governance (ACEA, 2010).

There is indeed a range of governance arrangements at different levels that potentially influences these systems. At the *international* level, there are agreements and regimes related to climate change mitigation, energy security, and innovation promotion. In the *EU*, the European Commission has its "Greening Transport Package" including a Communication (CEC, 2008), and a strategy for external cost integration. In 2011, the EU adopted a roadmap for the next decade to reduce its dependence on imported oil and to cut carbon emissions in transport by 60% by 2050 (CEC, 2011). By 2050, key goals include phasing out conventionally fuelled cars in cities, and a 50% shift of medium-distance intercity passenger and freight journeys from road to rail and waterborne transport. The EU has set progressive targets for carbon dioxide (CO_2) emissions from cars, and the current target is to reach an average of 130 g CO_2/km by 2015. The EU has also governed through various projects, for example, the 1990s ZEUS programme (Zero and Low Emission Vehicles in Urban Society); a collaborative effort of eight European cities to facilitate the introduction of cleaner cars. At the *national* level, governments use different instruments to facilitate the deployment of sustainable transport technologies, such as differential taxation/licensing fees of vehicles, differential taxation of fuels, investment subsidies, discharge premiums, and R&D programmes for alternative fuel and high-efficiency vehicles. Vehicle fuel standards and emissions standards have been introduced at both national and EU levels. The Californian Zero Emission Vehicle (ZEV) mandate and the US Corporate Average Fuel Economy standard (CAFE) are well-known examples that have affected the industry not only in America but also internationally. *Local* authorities have deployed measures such as differential parking fees and congestion charges, as well as developing infrastructure and refuelling/charging stations for vehicles.

Despite the wealth of governance to induce sustainable innovation in transport, there is surprisingly little analysis of it. Questions about *who* is best suited to govern sustainable innovation in transport, what are the appropriate *levels* for governance, and what are the most effective *instruments* to use, have not been addressed much at all. This lack of knowledge is not unique to the transport sector but concerns technological innovation much more broadly. Just how governance should be best arranged to achieve both momentum and sustainable direction in technological innovation systems is not well understood, be it in biotechnology, electricity generation and use, or urban infrastructure. Agreeing on governance can be complicated due to lack of agreement on goals, and divergence of interests between the many actors involved across levels. Still, there is, at least in the EU, relatively strong consensus among actors, judging from the plethora of low-carbon road maps and visions for sustainable energy and transport systems (Nilsson et al., 2011).

Existing research on governance for sustainable technologies has usually been national or sometimes regional in scope – even when technological innovation systems rather than regional or national innovation systems are in focus (Jacobsson and Lauber, 2006b). But sustainable transport technologies emerge internationally, are taken up on international markets, and respond to global challenges such as resource use and climate change. Furthermore, the technologies are put to use locally, and constrained or enabled by local rules and infrastructures.

This paper examines patterns of governance across these levels in two low-carbon vehicle technology domains, namely biofuels and hybrid-electric vehicles. The main research question is: in what ways are the characteristics of the governance arrangements different or similar across the two technology domains of biofuels and hybrid-electric vehicles, and how can we explain these similarities and differences? The paper takes a multi-level approach but the entry point for the national level is the country of Sweden. The purpose of the paper is to make an overall comparative assessment of governance patterns over the last two decades and make a classification of their characteristics. Here, we take a particular interest in understanding (1) actor involvement, for instance, the relative involvement of private and public actors; (2) the level of governance, and (3) the mechanism of governance.

Following the characterisation, we discuss how similarities and differences between the systems might be explained from different policy-theoretical lenses.

The paper unfolds as follows. Section 2 introduces the relevant debates on governance within and outside the innovation literature, and develops two main analytical categories for the empirical inquiry: "Who governs?" and "How to govern?" Our methodology is then described in Section 3. Section 4 introduces biofuel and HEV technology systems and maps related governance. Section 5 discusses similarities and differences and probes some tentative explanations for the observed patterns. Section 6 concludes the paper.

2. Towards governance mapping – relevant perspectives and debates

Governance is used throughout this paper as including public policy as well as underlying institutional arrangements and capacities for societal coordination. The interest in governance in both academia and the policy arena is connected to several debates. The first concerns whether processes of preparing, deciding on and implementing measures to advance societal objectives increasingly involve – or should involve – stakeholders other than the nation state (such as NGOs, corporations or regional organisations) (Pierre and Peters, 2006). The second concerns whether interactions between private and public actors have changed and hence the coordination necessary to achieve, for instance, sustainable innovations, relies more on measures other than control and coercion. The third concerns the political process, with questions about "good governance" such as transparency, accountability and democratic legitimacy (for instance regarding private sector powers and decision making) (Newell et al., 2008). These and other debates are then applied to different problems such as how to device institutional architectures (Biermann et al., 2009) or how to solve problems in a multi-level governance context (Hooghe and Marks, 2003). Thus, the governance debate is multifaceted and cannot be comprehensively unpacked and addressed in a single applied paper. In an attempt to rescue something distinctive for a framework useful for mapping governance, we draw on a selection from the broader governance literature and from some more specific literatures on technological innovation.

First, examining what specific literatures on technological innovation have to say about governance, a first perspective is based on the traditional "S-curve" for technology systems change and market development, which has been used as a basis for identifying different governance needs. For instance, as technologies move from pre-market development phases through to niche markets and scaling up, it has been suggested that policy should change (Azar and Sanden, 2011; IEA, 2011; Midttun and Gautesen, 2007). Early-stage technologies may need R&D investments, technology-specific support schemes, and protected spaces for experimentation and learning. In the more mature technology phase, governance based on broader market-efficiency appears more appropriate. In this phase, different technologies should compete on equal terms, and governance would preferably focus primarily on compensating for market failures.

A second tradition in technology innovation analysis stresses that a number of processes are on-going simultaneously in a technological innovation system and are subject to different barriers and enablers, including governance. These processes include knowledge development and diffusion, which can be enabled through R&D support, or market formation, facilitated by subsidies or procurement, but also "resource mobilisation", "entrepreneurial experimentation", and "legitimation" (Bergek et al., 2008; Hillman et al., 2011). These latter processes imply a wider variety of governance needs than traditional technology-support policies have tended to cover, including the formation of networks, government procurement, assured market sales and price fixing (see, e.g., Edquist et al., 2000; Jacobsson and Bergek, 2004; Nygaard, 2008; Jacobsson and Lauber, 2006a). In line with the governance perspective, this innovation-systems research also shows that non-state actors can be better positioned to initiate governance – preferably in cooperation with universities and local governments as suggested by the "triple-helix" metaphor (Etzkowitz, 2003).

A third approach is to analyse technological change through a multi-level framework centred on the socio-technical regime (Geels, 2004). Regimes can be thought of as highly interrelated institutional, technological and cognitive structures characterised by established products and production technologies, supporting infrastructures, stocks of knowledge, engineering and user practices, social relations, and networks, norms, institutions and regulations. Building on this frame, transition

management and strategic niche management are portrayed as governance in themselves (Kemp et al., 1998; Loorbach, 2010). They highlight in particular the governance needs for providing visions, building networks and platforms for stimulating learning, and making sure that a variety of options is explored (Dijk and Kemp, 2012). How well technologies align with regimes has implications for governance. A technology that is located outside the dominant regime may suffer a range of barriers and governance "deficits" (Markard and Truffer, 2008). Thus, technologies' isolation from the regime or protection from within the regime should be considered in designing effective governance.

The more generic literature on governance is both broad and ambiguous, but adds important insights. Although past attempts to conceptualise governance have often led to more confusion than clarity, better analytical clarity has been achieved in some recent contributions (Newell, 2008; Treib et al., 2007). These contributions articulate key dimensions, allowing the development of operational concepts that are needed to empirically map and classify governance over time. Treib et al. (2007) formulate three distinct and separate governance dimensions: the policy dimension (what instruments are used?), the politics dimension (who participates?), and the polity (institutional) dimension (what are the rules of interaction?). Newell et al. (2008) pose similar questions such as: what is to be governed, who is governing, on whose behalf, and how do they govern (through what instruments, mechanisms and rules)?

Combining these generic governance perspectives with the more specific governance aspects of the innovation and transition literatures, our framework for mapping governance distils two main dimensions and five core variables therein.

2.1. Who governs?

'Who governs?' examines, first, *whether the state, alone or in some combination with private actors, orchestrates the governance arrangement, and, second, at what level of governance the arrangements are enacted, from the local to the international.*

The actor variable figures centrally in governance debates (Kohler-Koch and Eisling, 1999). Descriptive models range from state-centric to society-centred ones. The state-centric ones focus on public actors and their institutional and political capacities to steer, whereas the society-centred approaches are concerned with a wider range of actors, including private sector ones, and their ability to self-organize in networks (Peters, 2000). The literature has predominantly dealt with state-centric approaches and the ways in which public and private actors are interacting, but governance is also an established concept in organisational and firm studies. Dyer (1996) suggested that within the corporate value chain structures in the early 1990s, Japanese alliance-based governance was more adaptive and more competitive than the US car manufacturers' hierarchical and vertically integrated system.

The level variable concerns both how much coordination takes place between actors and networks at local, regional, national, international and global levels, but also the interaction between levels and scales, be they geographical or organisational (Hooghe and Marks, 2003). Intuitively, the existence of multilevel governance in any policy domain appears to be a fact rather than a question, in the sense that it is obvious that governance occurs within and across levels. Still, scholars are unclear about whether there are trends and if so in what directions, and what role there might be for governance at different levels in enabling sustainable transport. The debate has been of particular concern in the European Union. The EU aspires to far-reaching coordination, both across sectors and at multiple levels but it is unclear whether EU institutions have been increasingly influential on national-level governance systems (Jordan and Schout, 2006; Knill and Lenschow, 2005). Transport policy is a case in point, where the EU takes an increasing initiative but where national and local levels dominate actual investments decisions.

Another debate concerns the alleged dissolution of the nation state as the major agent in governance – through processes of globalisation, regionalisation and localisation. Kaiser and Prange (2004) argue that such trends are evident in the field of innovation, and that regulatory activities have gradually moved away from national states to organisations at multilateral and regional levels. Indeed, the entire perspective of regional innovation systems aligns with this view (Asheim and Coenen, 2006).

2.2. How to govern?

'How to govern?' examines, first, the mechanism of governance, i.e. whether regulative, market-based, normative or cognitive modes are deployed, second whether governance is on the supply (push) or on the demand (pull) side, and third, the technology specificity – whether it covers the whole sector (e.g., personal vehicles), the technology as a whole (e.g., hybrid-electric vehicle technologies), or a specific subsystem (such as battery/energy storage, electric machinery, or power electronics.)

We base the mechanisms variable on the well-established 'hierarchy-market-networks' typology (Peters, 1998; Scharpf, 1993). Hierarchical approaches tend to rely on *regulative* mechanisms such as enforcing certain standards and procedures. Market approaches rely on market-based mechanisms, influencing the economic incentives and behaviour of market actors. Networked approaches tend to rely on cognitive or normative mechanisms. *Cognitive* mechanisms relate to the development of knowledge and the 'better argument'. This cognitive mechanism is particularly interesting in the field of innovation, in itself a knowledge-development process. *Normative* mechanisms relate to development of norms, preferences and social values. Norms may relate to processes (how things should be done, or 'logic of appropriateness' (March and Olsen, 1989; Scott, 1995) but also goals for policy and society. Network-oriented governance includes cooperative procedures such as private–public partnerships.

In their nomenclature of governance models, Pierre and Peters (2006) associate the characterisation of mechanisms with specific actor configurations. However, a certain actor constellation does not necessarily lead to a certain set of mechanisms. A state or a private firm can deploy a range of different mechanisms (Treib et al., 2007). For example, although regulatory standards and economic instruments have been the traditional means of governance for the state, they are also increasingly used in the private sector (e.g. procurement standards).

Strategies geared towards technology push versus market pull and degree of technology specificity are additional themes examined. On the former, recent EU policy efforts have shown increasing interest in market-pull complementing the traditional push-through-RD&D axiom (CEC, 2007) but it is unclear to what extent this signifies a trend shift. On the latter, technology neutrality is often heralded as a guiding principle by economists and the mainstream of policy makers, whereas innovation scholars increasingly have come to emphasise the need to make explicit choices on technological strategies, and the importance of technology-specific support (Azar and Sanden, 2011).

3. Methodology

We address the "Who governs" dimension by distinguishing, first, three types of private–public actor involvement; no/little, moderate, or high involvement of private actors. Second, we distinguish three levels of governance, local/regional, national and EU/federal. We use Sweden as the national jurisdiction and entry point to the analysis, but do not account separately for every local initiative at the city level (there are more than 200 municipalities in Sweden, many of them with some sort of sustainable transport initiative). For the "How to govern" dimension, we distinguish four mechanisms: regulative, market-based, normative or cognitive; three targets: push, pull, or both; and three levels of specificity: sector-wide (fuels or vehicles), system (biofuels or HEVs) or subsystem (ethanol, biogas, batteries, electrical propulsion systems). Sector-wide approaches are not necessarily the same in our two cases as they cover adjacent sectors (see below).

The study only includes governance with public involvement, but in principle public actors need not be involved in all governance. This distinction is however necessary to demarcate governance from, for example, corporate strategies. The identification and classification of governance arrangements (see Appendices A and B) were based on interviews with representatives for key actors in the two fields, as well as documents issued by national and international regulators and governmental bodies, firms and NGOs. The studied documents include policy documents, project and annual reports, public evaluations of state-funded R&D and demonstration programs, reports from financial press, automotive press and press releases from automotive manufacturers. This was complemented with a wide sample of interviews with key persons, who confirmed and added activities and governance arrangements, and pointed out relevant documents. We carefully selected respondents to cover the

development throughout the whole epoch, and made sure that different types of actors in the system were represented. In terms of larger programmes involving many projects, research projects with a few university researchers were not counted individually (Hillman, 2011; Magnusson, 2011).

The distinctions and scorings of different levels of, for instance, private–public involvement, were made through a qualitative–interpretative group exercise. An individual qualitative assessment was followed by a moderated group discussion to arrive at an agreed estimate of the relevant characterisation. While certain quantitative measures could also be used, e.g. volume of money, number of publications, number of people engaged, emission reduction achieved, etc., none of these indicators are useful (or even meaningful) for all types of governance. One solution would be to use several indicators to analyse the material, but this would not solve the problem of comparing different indicators.

Following an innovation systems approach (Carlsson and Stankiewicz, 1991; Bergek et al., 2008; Hillman et al., 2011), we defined system boundaries in terms of the actors, networks and institutions involved in developing the focal technologies. We used the products based on these technologies as focusing devices for the study. The systems definition also involved a spatial delimitation, restricting systems development to Sweden, as well as a temporal delimitation, covering the 1990–2010 period.

That system boundaries are tricky to establish is a well-known concern in innovation systems analysis. The firms that are active in innovation and technology development may have local or national bases for R&D or production. However, they tend to operate on international markets and are therefore affected by governance on other important markets. Second, also within and between companies, significant collaborations and joint projects are carried out across the world, which means that a German or Swedish automotive manufacturer may (and do) collaborate with Japanese ones on R&D and components, and thus may be affected by Japanese R&D policies. Interviews helped us to identify what foreign market governance to include. The delimitations made for geographical, temporal or technological scope and level of aggregation reduced the volume to a manageable set of observations, without sacrificing much understanding of the case.

4. Empirical analysis: governance for biofuels and hybrid-electric vehicles

4.1. Introducing the two technology systems

Biofuels cover several technologies; their value chains all starting with a biological material (called biomass), and the end product being used for propulsion of vehicles. We focus on four biofuels technologies defined by their respective value chains: biodiesel, biogas, ethanol and synthetic fuels (see Fig. 1).[1] These alternatives are often divided into a first and a second generation, largely reflecting whether they are commercially available (unbroken lines in Fig. 1) or under development (dashed lines), respectively. With this division, all of the synthetic fuels, as well as cellulosic ethanol, are referred to as second generation.

Today, only first-generation biofuels have reached the market. This, on the other hand, has become a truly global market with major producers including USA (maize ethanol) and Brazil (sugar-cane ethanol). Global biofuel production took off in the 2000s, growing from 16 billion litres in 2000 to more than 100 billion litres in 2010, and now accounting for 3% of the total road transport energy (IEA, 2010). The EU has set a binding 10% target on biofuels replacing petrol and diesel for transport by 2020 (European Parliament and Council of the European Union, 2009). In Sweden, particularly the use of ethanol, RME (rapeseed methyl ester) and biogas increased from almost zero in the 1990s to 5.7% of the energy used for road transport in 2010 (Energimyndigheten, 2011). Ethanol is either blended by 5% into conventional petrol or used in E85 (85% ethanol, 15% petrol), aimed for the increasing fleet of flexifuel cars, and E95 for heavy vehicles. RME is primarily blended into conventional diesel, while biogas is used in its pure form for cars and heavy vehicles.

[1] Here we leave out hydrogen and electricity, which could also be produced from biomass. These can be viewed as separate systems, with ongoing governance.

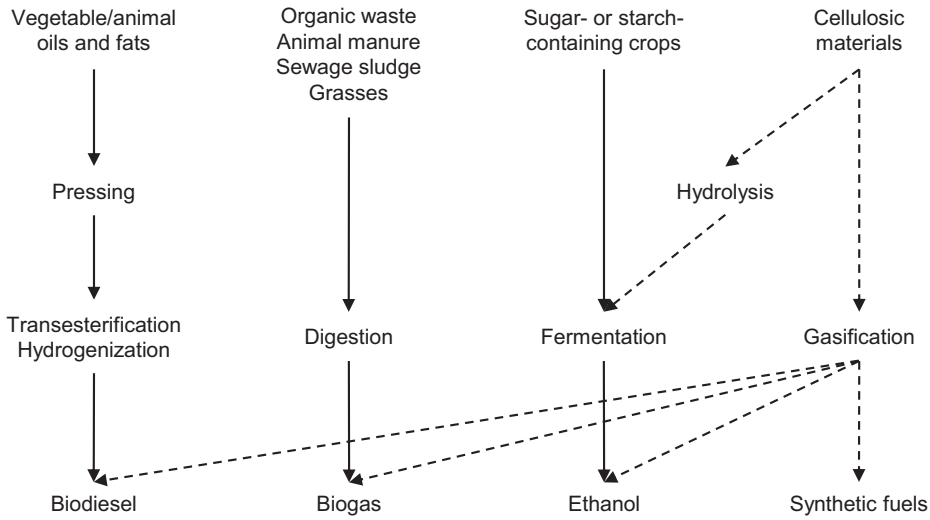

Fig. 1. Biofuels and their respective value chains, i.e., raw materials and production processes (modified, Hillman, 2008). Unbroken lines indicate existing paths, while dashed lines represent paths that are under development. Oils and fats, sugar and starch, and cellulosic materials may also be derived from residues and waste.

Though the introduction of biofuels in Sweden on environmental grounds was fairly unique, there have been some injections from other countries. First, the fuel ethanol programme in Brazil that started in 1975 showed that it was possible to run vehicles on high-level ethanol blends on a large scale. Later, Brazil became the major supplier of ethanol for the Swedish market. Second, flexifuel cars that can run on any mix of petrol and E85 were sold in the US since the 1990s, though they were scarcely fuelled with ethanol. Third, natural gas vehicles and related fuels were marketed in several countries, including Sweden. These vehicles could in principle run on biogas instead of fossil natural gas.

HEVs combine ICE engines with electric motors. They can be configured in a number of ways (Chau and Wong, 2002; Ehsani et al., 2005). One means of classification refers to the extent to which the vehicle relies on the combustion engine and the electric motor respectively for propulsion. At one end of this spectrum is the traditional vehicle, which only uses a combustion engine, and at the other end of this spectrum is the pure electric vehicle (EV), which only relies on an electric motor. In general, the potential for fuel efficiency gains becomes higher with a higher degree of hybridisation. *Mild hybrids* comprise relatively small electric motors, which primarily assist the combustion engine during acceleration from standing still and at low speeds. *Full hybrids* (e.g. Toyota Prius) comprise a high voltage electric system, and a more powerful electric motor which is integrated in the automotive power train to assist the combustion engine at both lower and higher speeds. The next step is the *plug-in hybrid*, which comprises a larger battery that may be charged via the electric grid. The electric motor is the prime source of motive power in the plug-in hybrid. The internal combustion engine allows battery charging while driving. The hybrid system may also be configured so that the engine may drive the car directly as the battery runs out of electric power. This enables longer-range drives. Hence, the plug-in hybrid can be viewed as an electric car with an on-board charging device. Since mild and full hybrids rely on the same fuelling infrastructure as traditional vehicles, they do not require any substantial behavioural changes. By contrast, plug-in hybrids require electric charging. This means that the user has to plan driving according to the time required for charging and the availability of electric charging stations.

Combustion engines and transmissions are based on entirely different knowledge bases than the knowledge needed for developing subsystems such as high-voltage electrical machines, energy storage

systems (batteries), as well as power electronics and energy management systems, which are needed in hybrid-electric power trains. In order to realize the potential gains in terms of fuel efficiency in a hybrid power train, these technologies need to be effectively combined into a coherent and optimised system. The complexity of the system and the large number of possible configurations means that engineers involved in hybrid technology development have to match different systems-level requirements, such as drivability, fuel efficiency, safety, reliability, performance, etc. Hence, hybrid-electric vehicle technology is very much about systems integration, combining different sets of engineering knowledge.

The first mass produced hybrid-electric car, the full hybrid Toyota Prius, was introduced in Japan in late 1997 and in export markets 2000–2001. Annual sales volumes of hybrids during the initial years were low; below 100,000 vehicles globally and below 50,000 vehicles in USA, which has been the largest national market for hybrids during the 2000s. However, in 2004, as Toyota released the second generation of their Prius model, sales on the US market started to soar. Between the years 2003 and 2007, there was a seven-fold increase of hybrid vehicle sales in USA reaching the top mark of 352,274 hybrid vehicles sold in 2007. In 2008–2010, the financial crisis struck the automotive industry and vehicle sales plummeted. In 2010, hybrids had a 2.3% market share in USA. In Europe, hybrid vehicle sales have been significantly lower with a market share consistently below 1%. An important explanation to this difference is that diesel cars maintain a significant share of vehicle sales in Europe. Since diesel engines are more fuel efficient than gasoline engines, gasoline-fuelled hybrid cars have faced tougher competition on European markets (Magnusson and Berggren, 2011). By 2011, all major automakers had hybrid vehicles in their model programmes, mostly full hybrids. Only one manufacturer had introduced a plug-in hybrid model – GM's Chevrolet Volt. However, several manufacturers had announced plans for introducing plug-in hybrids over the coming few years.

4.2. Comparing the two technology systems

When comparing the two systems up front, two distinctive differences appear.

First, the two fields have reached different levels of market uptake and diffusion in Sweden. Although predevelopment has gone on for a long time, the HEV system is now in an acceleration phase. Toyota Prius, market leader on HEV, had sold cumulatively 500,000 cars in 2005, crossed 1 million in 2008, and in September 2010 it reached 2 million. Biofuels have over the last decade established a significant market presence, especially in Sweden. Fig. 2 displays the market uptake in Sweden of the two technologies in terms of clean car sales from 2004, when the clean car category was established, showing how biofuels dominated 2004–2008. A marked decline occurred in 2008 due to changes in support schemes, when small and efficient diesel cars were given "clean car" status and associated market support. By comparison, hybrid car sales have consistently been very limited during the studied period.

The market uptake is of course strongly related to the relative costs of technology. Since the hybrid vehicles comprise a lot of additional and expensive subsystems (including batteries) compared to conventional vehicles, attaining competitive cost levels is a prime challenge for HEV manufacturers (Chanaron and Teske, 2007).

Second, the two fields link to different adjacent sectors. The HEV system links the automotive sector to electrical propulsion systems, power electronics (i.e. semiconductors) and battery industries in new business constellations and cooperative arrangements. The biofuels system links the automotive sector to a partly new fuel supply chain. Although the conventional fuel companies are engaged, there is also a range of new commercial actors involved, including from the agriculture and forest sectors.

At the same time, the systems share important features. First, both systems, as part of the road transport sector, have gone through turbulent times with a global car manufacturing crisis, restructuring processes and rapidly shifting governmental policies and consumer demands. These developments take place at and across international, national and local levels. Arguably, the car industry is in the midst of a transformational change as a result of global market and policy conditions (Mikler, 2005). Second, generic policy objectives, strategies and instruments at different levels influence both systems, including transport, climate, energy and innovation policies.

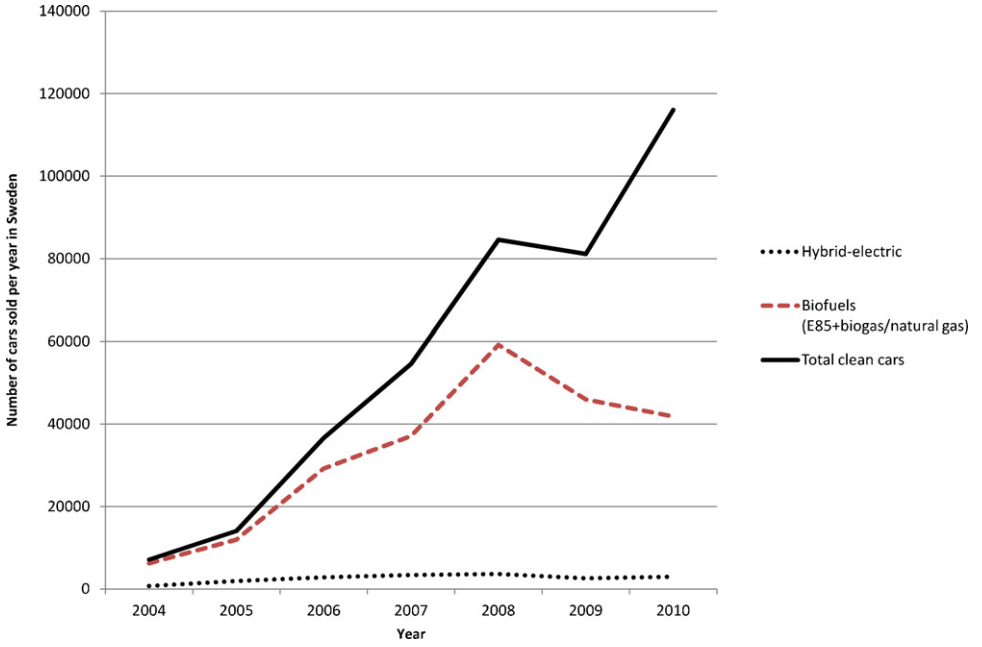

Fig. 2. Clean car sales in Sweden 2004–2010.
Source: www.bilsweden.se.

In the following we will look more closely at specific governance patterns from international to local levels (and we use Sweden as entry point at the national level).

4.3. Mapping of biofuels governance

For biofuels, three major groups of governance can be discerned. First, throughout the studied period, a range of governance arrangements influenced the focus of technology towards and within the field of biofuels. The most influential ones, according to our survey, were the national energy policy agreement in 1991, a national biofuels introduction strategy of 2001–2002 by four governmental agencies', and a number of EU policies after 2001. The agreement and the strategy favoured ethanol and synthetic fuels, respectively, while EU directives set targets for the market introduction of biofuels in general. Second, a large share of the research, development and demonstration of biofuels and vehicles was part of governance, i.e., programmes with a certain degree of public involvement from various levels (mainly national and EU), for example regarding initiation, coordination and funding. Normally, such programmes also entailed a moderate or high degree of private participation and co-funding. Third, the market development for first generation biofuels and vehicles largely hinged on an increasing number of arrangements from around 2000. This entailed public procurements of buses and cars, and various economic incentives and regulations related to biofuels and clean cars. Incentives were initiated on both the national level (e.g., tax benefits, clean car premium) and the local level (free parking, car subsidies).

In summary, the shares of governance arrangements that were allocated to each alternative for each variable are shown in Fig. 3 below (see Appendix A for details). On 'Who governs?', most arrangements were implemented on a national level with no or little involvement of private actors. The picture is more fragmented when it comes to 'How to govern?' with both push and pull arrangements working through cognitive, market and regulative mechanisms and with both system and subsystem levels commonly used.

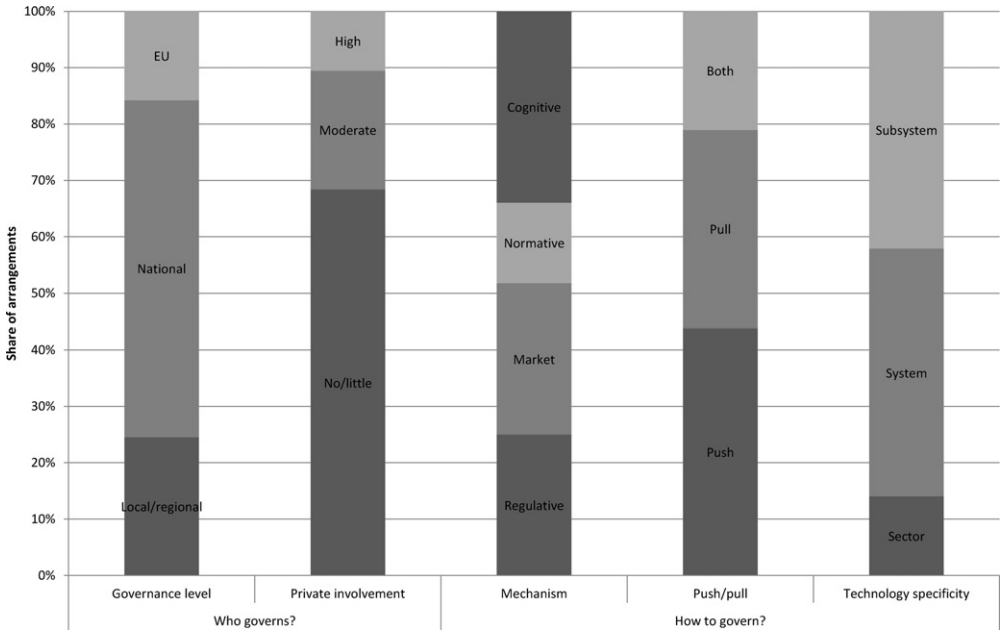

Fig. 3. The shares of biofuels-related governance allocated to different categories.

4.4. Mapping of HEV governance

Turning to HEV, there are both similarities and differences in the governance architecture. Similar to biofuels, most of the influential governance arrangements have been implemented on a national level. However, the degree of private involvement is significantly higher in the case of hybrid-electric vehicles, especially during the 2000s. In particular, this points to the central role that the established vehicle manufacturers have in the development of hybrid-electric vehicle technology. Entry barriers are high in the automotive industry and only the established manufacturers possess the required resources and system integration capabilities to develop and produce hybrid-electric vehicles. However, these manufacturers also need to acquire knowledge in new fields of technology to accomplish this (e.g. battery technology, electric machinery, and power electronics). State-funded R&D programs promote this. During the 2000s, nearly all these programs were designed as public–private partnerships, involving both automotive manufacturers (and supplier organisations) as well as technical universities. Apart from these cognitive mechanisms, regulative mechanisms in terms of various emissions regulations on different levels have also been influential in the HEV system. Whereas both cognitive and regulative mechanisms have been influential in both our cases, the HEV case is more centered on cognitive and regulative mechanisms. The focus on R&D (technology development, prototyping, testing, field trials, demonstrations, etc.) in HEV governance also results in a high degree of technology push. The market pull arrangements mainly refer to various forms of subsidies for low emitting vehicles. Finally, the HEV case is characterised by a relatively low degree of technology specificity. The main part of the identified governance arrangements addresses the sector-wide level, the automotive sector. Again this points to the dominant position of the established vehicle manufacturers. Addressing the sector in general, rather than a particular system or subsystem, means that the manufacturers have a large influence in the selection of technologies.

In summary, the shares of governance arrangements that were allocated to each alternative for each variable are shown in Fig. 4 below (see Appendix B for details). On 'Who governs?' most arrangements

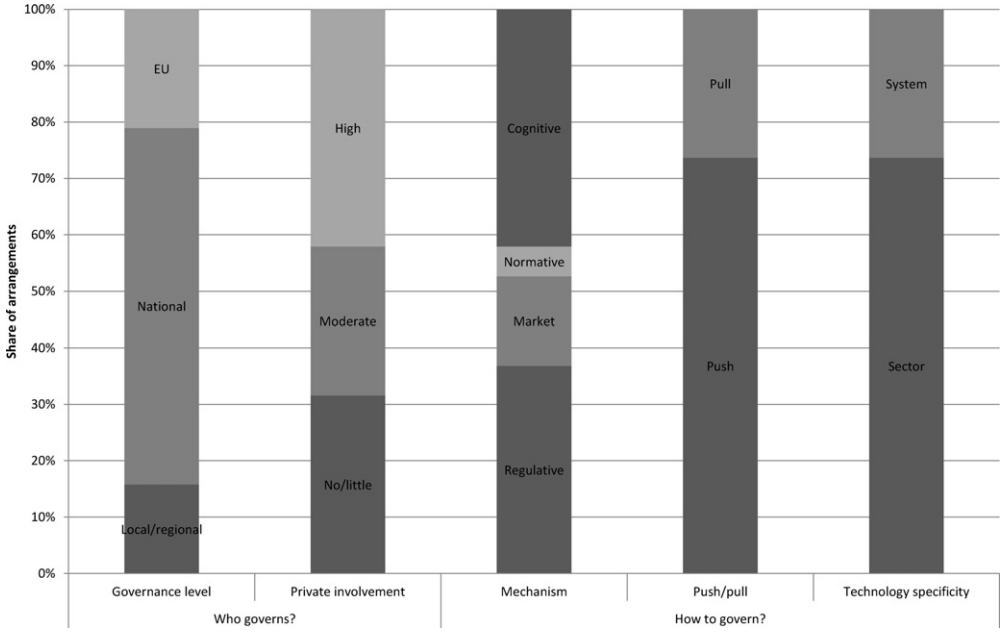

Fig. 4. The shares of HEV-relevant governance allocated to different categories.

were implemented on a national level with significant involvement of private actors. On 'How to govern?', the cognitive mechanisms are markedly stronger and the market mechanism weaker than in biofuels, with a focus on technology-push arrangements, and the sector-wide level most commonly used.

5. Discussion

How can we understand these differences? In the following section we first summarise governance differences and similarities between the two systems, and then offer a tentative set of explanations for these differences, without aspiring to a solid explanatory analysis.

5.1. Differences and similarities between HEV and biofuels governance

The two systems address partly different technology types, one focussing on an engine technology and the other on a fuel. Still, HEV and biofuels both offer partial solutions to the challenge of reducing the climate impact from the transport sector. Both systems are built on well-known technologies, researched and developed over at least 40 years. Both are tightly connected to the same global industrial field – the automotive sector. Both are subject to the same transformational pressures from global market conditions and from international climate policy. One could therefore expect, from a rational policy perspective, similarities in how the two systems are governed. Still, summing up and comparing the systems empirically, there are marked differences in governance patterns across them.

Looking at the "Who?" dimension, and the roles of the public and private sector, it is clear that biofuels has a stronger public bias, and probably even has developed a significant dependency on governmental support, whereas HEV is to a much greater extent governed by industry. On levels of governance, there are no strong differences: international-level governance is weaker in both but more initiatives are found in biofuels. National-level governance in *other jurisdictions* is important for HEV – e.g. the ZEV mandate. Many governance arrangements function where the market is, and

what happens in terms of Swedish governance is therefore less relevant from the perspective of the automotive manufacturer. Some limited international governance in HEV has been orchestrated by for instance IEA and by various interest organisations with more or less industrial focus. However, the impacts of such governance have not been possible to discern, as it focuses primarily on monitoring and reporting of progress. Much of the coordination work is handled by the manufacturers themselves, with limited state involvement, and this coordination work naturally transcends national boundaries.

Comparing the "How?" dimension, biofuels exhibit a wide array of governance, fairly balanced across market, normative, regulative and cognitive mechanisms. In contrast, market support has not been implemented specifically for HEV until 2012, when Sweden implemented a subsidy for "super clean cars" (such as plug-in hybrids). Instead, cognitive governance dominates as far as specific measures are concerned. However, the HEV system is of course influenced by sector wide arrangements also of regulatory and market type. Here we see the most notable difference, which is the extent to which biofuels governance has been geared towards very specific subsystem technologies such as ethanol, whereas the HEV system has been predominantly shaped by sector-wide approaches. Finally, HEV has been subject to a larger emphasis of technology push over market pull arrangements.

It can be noted that, using the same mapping approach, we identified nearly three times more arrangements in biofuels than in HEV. The number of arrangements is a rough indicator that has some primary disadvantages, such as not accounting for the pervasiveness of the various arrangements, and not differing between large and small R&D programmes or economic instruments. Still, the difference is so large that it signals a differentiated level of initiative from the state.

5.2. How can we understand the differences in governance patterns?

From a socio-technical perspective, the positioning of the technology within or outside the socio-technical regime is a possible explanation for governance patterns. Biofuels, although representing change in terms of actors and technologies, are easily accommodated within the existing personal automotive system, in terms of vehicle design, travel behaviour and supporting infrastructure. It does not require automotive manufacturers to redesign vehicles, to make connections or collaborations with new industries, or to significantly depart from their existing business models. HEV, on the other hand, partly operates outside the automotive regime. There are aspects of regime-level changes in infrastructure, ownership models, and driving behaviours (although they are more pronounced in the case of full EVs). Main problems relate to access to knowledge and suppliers within new fields such as batteries and power electronics, how to integrate this knowledge and supply chains in existing production systems, and new requirements for industry standards and regulations, e.g. for charging and safety at accidents. This differential challenge is reflected in the types of governance, which in the case of HEV focus much more on cognitive modes and network building – key processes when regime-external processes are developing.

From a political perspective, a powerful predictor of policy interventions is to examine who are the most influential actors and actor networks (Baumgartner, 2007). Can we find systematic differences, at the national level, of interest groups, industries and others, that benefit from or lose out from governance measures? HEVs represent changes in the vehicle supply chain, towards batteries, power electronics and electric propulsion systems. New suppliers are required, but R&D and production are still orchestrated by the automotive manufacturers. Retaining central positions as systems integrators, the established automotive manufacturers have thus had a significant influence on governance. Biofuels, on the other hand, represent a change in the fuel supply chain, where new powerful actors such as the agricultural sector come into the picture. This might explain the greater focus on technology specificity in biofuels, as a variety of influential actors with their preferred technological solutions have been able to influence how governance is arranged.

From an institutional perspective, a possible explanation for differential governance, particular in terms of who governs, can be derived from theories of institutional path dependencies (Steinmo et al., 1992) and policy traditions (Howlett, 1991). According to this perspective historical legacies of previous decisions form the institutional basis for current and future governance patterns.

Some sectors may have a stronger tradition of operating on the political arena, and demanding support and initiatives from the government, including market support. Here, we posit that there are policy traditions and legacies in the actual degree of public interest in the systems that characterise the biofuels (and oil) system. Car fuels in particular is traditionally considered a public affair and a matter of political debate, whereas HEV subsystems such as electronics and batteries have traditionally been purely industrial concerns, with limited expectations on, and interest in, public policy intervention. As a result, industry has a stronger role in the case of HEV.

Linked to this, theory on socio-technical regimes would suggest that technology interacts with governance through positive feed-back in both directions, so that as technologies get introduced, driven by governance, they in turn ignite the emergence of new governance arrangements – leading to the institutionalisation and self-reinforcement of a regime. This would explain the much greater volume of arrangements in biofuels: the biofuels penetration on the market is much stronger – driven by public support schemes, but also driving governance such as sustainability criteria and other standards. However, our data does not allow us to validate this claim.

6. Concluding remarks

The two cases examined display systematic differences in governance, with a much stronger public engagement in biofuels than in HEVs. However, although the uptake of HEVs has lagged behind the uptake of biofuel vehicles in Sweden during the past 10 years, the public engagement in biofuels cannot be seen as a panacea. It may also have created a system that is more dependent on government support. To what extent the biofuels acceleration will be a lasting effect is still to be seen, in particular since governments' enthusiasm for them has cooled off significantly in recent years. From 2012, in Sweden, ethanol cars will no longer automatically qualify for clean-car subsidy schemes. As a result, to what extent biofuel vehicles will be able to compete as low carbon alternatives in the future is increasingly called into question, and likely depends on the commercialisation of second-generation biofuels.

At the same time, one could argue that the more transformative innovation process inherent in a move towards plug-in hybrids and electric vehicles (but not mild and full hybrids), including changing user behaviours, infrastructures, as well as vehicle designs, would require more active governance of a broader palette than has so far been realised.

Although the EU has been active in recent years, for instance through R&D support and sustainability criteria for biofuels, international governance is largely missing when it comes to innovation in sustainable transport technology. Is there an unfulfilled potential? This is an open question still. On the one hand, the politically uncoordinated industrial system has in many cases proven very effective in generating innovation and experimentation, adapted to different conditions and contexts shaped by national policies and regulations. On the other hand, actors do call out for international harmonisation of rules and standards that are in line with the markets on which the industrial actors are operating.

The need for governance at different levels may in the end depend not primarily on who governs, but rather who is being governed, i.e. what actors are driving the technological development forward? Automotive manufacturers need long term and coordinated steering to motivate the necessary investments for developing new power trains. At the same time, only existing large actors have the resources and capacities needed to do just that.

This paper has had predominantly conceptual and descriptive ambitions, in order to operationalize, categorize and map governance. Therefore, our explanatory discussion is only tentative, but the connections to theories of socio-technical regimes and transitions, interest group influence and institutionalism are clear and show potential for further research.

Acknowledgments

This research was funded by the Swedish Research Council FORMAS. The authors would like to thank Annika Rickne as well as two anonymous reviewers for helpful comments on earlier drafts.

Appendix A. Classification of biofuel governance

Arrangement	Start year	End year	Governance level	Private involvement	Mechanism	Push/pull	Technology specificity
Stockholm ethanol bus project	1990	1993	Local/regional	Moderate	Cognitive	Push	Subsystem
Foundation for Swedish Ethanol Development (SSEU)	1990	1999	National	Moderate	Normative	Both	Subsystem
Biogas in Linköping	1990	Ongoing	Local/regional	No/little	Cognitive	Push	Subsystem
Three party agreement	1991	1991	National	No/little	Regulative	Push	System
Biofuels Programme	1991	1997	National	Moderate	Cognitive	Both	System
Biogas and ethanol facilities	1992	Ongoing	Local/regional	No/little	Market	Push	Subsystem
Ethanol and biogas bus purchase	1992	Ongoing	Local/regional	No/little	Market	Push	Subsystem
Ethanol Development Programme	1993	1997	National	No/little	Cognitive	Push	Subsystem
Flexifuel car demonstration	1994	1996	Local/regional	Moderate	Normative	Push	Subsystem
Commission on Alternative Fuels (SOU 1996:184)	1995	1996	National	No/little	Cognitive	Push	System
Clean vehicle organisations	1995	Ongoing	Local/regional	No/little	Normative	Pull	System
Fuel tax exemptions	1995	Ongoing	National	No/little	Market	Pull	System
KomKom ethanol introduction strategy (SOU 1997:35)	1996	1996	National	No/little	Cognitive	Both	Subsystem
Various EU programmes	1996	Ongoing	EU/federal	Moderate	Cognitive	Both	System
Biogas as a Vehicle Fuel (SOU 1998:157)	1998	1998	National	No/little	Cognitive	Push	Subsystem
Local Investment Programme (LIP)	1998	2002	National	No/little	Market	Pull	Sector
Ethanol from Forest Raw Material	1998	2004	National	No/little	Cognitive	Push	Subsystem
Clean car sharing	1998	Ongoing	local/regional	Moderate	Market	Push	System
Clean vehicle procurements	1998	Ongoing	local/regional	No/little	Market	Pull	System
Municipal clean vehicle incentives	1998	Ongoing	local/regional	No/little	Market	Pull	System
Parking fee subsidies	1998	Ongoing	local/regional	No/little	Market	Pull	System
SRA travel policy	1998	Ongoing	National	No/little	Market	Pull	Sector
BioAlcohol Fuel Foundation (BAFF)	1999	Ongoing	National	Moderate	Normative	Both	Subsystem
Fringe benefit tax reduction	1999	Ongoing	National	No/little	Market	Pull	System
Local clean car subsidies	2000	Ongoing	local/regional	No/little	Market	Pull	System
Commission Communication on Alternative Fuels	2001	2001	EU/federal	No/little	Normative	Both	System
Introduction of Biofuels on the Market	2001	2002	National	No/little	Normative	Push	System
Biogas in Vehicles	2001	2004	National	Moderate	Cognitive	Push	Subsystem
Biogas West	2001	Ongoing	local/regional	High	Market	Push	Subsystem
Black liquor & DME	2001	Ongoing	National	High	Cognitive	Push	Subsystem
EU Biofuels Directive	2003	2003	EU/federal	No/little	Regulative	Both	System
Commission on Renewable Transport Fuels	2003	2004	National	No/little	Cognitive	Both	System

Appendix A (*Continued*)

Arrangement	Start year	End year	Governance level	Private involvement	Mechanism	Push/pull	Technology specificity
FALT (Research on Alternative Transport Fuels)	2003	2006	National	Moderate	Cognitive	Push	Subsystem
Climate Investment Programme (Klimp)	2003	2008	National	No/little	Regulative	Pull	Sector
National definition of clean cars	2004	2005	National	No/little	Regulative	Pull	Sector
Chrisgas pilot plant	2004	2009	National	Moderate	Cognitive	Push	Subsystem
Ethanol Bus (and Truck) Initiative	2004	Ongoing	National	No/little	Market	Pull	Subsystem
Ethanol pilot plant	2004	Ongoing	National	No/little	Cognitive	Push	Subsystem
Public procurement of clean cars	2004	Ongoing	National	No/little	Regulative	Pull	System
Saab BioPower release	2005	2005	National	High	Normative	Push	Subsystem
Sekab purchase	2005	2010	local/regional	High	Several	Push	Subsystem
Diesel standard adjustment	2006	2006	National	No/little	Regulative	Push	Subsystem
EU Strategy for Biofuels	2006	2006	EU/federal	No/little	Cognitive	Both	System
Congestion tax exemption	2006	2008	local/regional	No/little	Regulative	Pull	System
European Biofuels Technology Platform	2006	2010	EU/federal	High	Cognitive	Both	System
GoBiGas	2006	2010	National	High	Cognitive	Push	Subsystem
Renewable fuel obligation	2006	2010	National	No/little	Regulative	Push	System
Clean car premium	2007	2009	National	No/little	Market	Pull	System
Ethanol energy combine	2008	2010	National	Moderate	Cognitive	Push	Sector
Nordic Ecolabel Certification	2008	2010	National	Moderate	Market	Pull	Subsystem
Vehicle conversion law	2008	2010	National	No/little	Regulative	Pull	System
EU RES Directive	2009	2009	EU/federal	No/little	Regulative	Both	System
EU CO_2 from cars	2009	2010	EU/federal	No/little	Regulative	Push	System
Vehicle tax exemption	2009	2010	National	No/little	Regulative	Pull	Subsystem
Biofuels sustainability criteria	2010	2010	EU/federal	No/little	Regulative	Pull	System
National biogas strategy	2010	2010	National	No/little	Normative	Both	Subsystem

Appendix B. Classification of HEV governance

Arrangement	Start year	End year	Level	Private involvement	Mechanism	Push/pull	Specificity
US Corporate Average Fuel Efficiency (CAFE)	1975	Ongoing	EU/federal	No/little	Regulative	Push	Sector
Californian ZEV-mandate	1990	Ongoing	National	No/little	Regulative	Push	Sector
Local exhaust emission regulation	1990	Ongoing	Local/regional	No/little	Regulative	Push	Sector
Local noise emission regulation	1990	Ongoing	Local/regional	No/little	Regulative	Push	Sector
US (EPA) Emission standards	1991	Ongoing	EU/federal	No/little	Regulative	Push	Sector
European (Euro) emission standards (NOx, PM, VOC)	1993	Ongoing	EU/federal	Moderate	Regulative	Push	Sector
KFB program	1993	2000	National	Moderate	Cognitive	Pull	System
Nutek R&D support	1993	1996	National	High	Cognitive	Push	System
Nutek public procurement	1993	1996	National	No/little	Market	Pull	System

Appendix B (*Continued*)

Arrangement	Start year	End year	Level	Private involvement	Mechanism	Push/pull	Specificity
EU R&D support	1994	Ongoing	EU/federal	High	Cognitive	Push	Sector
Subsidies and tax incentives for low CO_2-cars	Diff	Ongoing	National	Moderate	Market	Pull	Sector
Green car 1	2000	2007	National	High	Cognitive	Push	Sector
Green car 2	2006	2008	National	High	Cognitive	Push	Sector
Swedish hybrid vehicle center	2007	Ongoing	National	High	Cognitive	Push	System
European proposed directive on CO_2-emissions	2007	Ongoing	National	Moderate	Regulative	Push	Sector
Transport for London	2007	Ongoing	local/regional	High	Normative	Pull	Sector
Energimyndigheten hybrid R&D support	2007	2008	National	High	Cognitive	Push	System
FFI	2009	Ongoing	National	High	Cognitive	Push	Sector
Scrappage schemes	2009	Ongoing	National	Moderate	Market	Pull	Sector
EU CO_2 from cars	2009	2010	EU/federal	No/little	Regulative	Push	System

References

ACEA, 2010. ACEA Communication – Vehicle and Biofuels towards 2020. European Automobile Manufacturers' Association, Brussels.

Asheim, B., Coenen, L., 2006. Contextualising regional innovation systems in a globalising learning economy: on knowledge bases and institutional frameworks. Journal of Technology Transfer 31, 163–173.

Azar, C., Sanden, B., 2011. The elusive quest for technology-neutral policies. Environmental Innovation and Societal Transitions 1, 135–139.

Baumgartner, F., 2007. EU Lobbying: a view from the US. Journal of European Public Policy 14, 482–488.

Bergek, A., Jacobsson, S., Carlsson, B., Lindmark, S., Rickne, A., 2008. Analyzing the functional dynamics of technological innovation systems. A scheme of analysis. Research Policy 37, 407–429.

Biermann, F., Pattberg, P., van Asselt, H., Zelli, F., 2009. The fragmentation of global governance architectures: a framework for analysis. Global Environmental Politics 9, 14–40.

Carlsson, B., Stankiewicz, R., 1991. On the Nature, Function, and Composition of Technological systems. Journal of Evolutionary Economics 1 (2), 93–118.

CEC, 2007. Com 2007/860 A Lead Market for Europe. European Commission, Brussels.

CEC, 2008. Communication: Greening Transport [COM(2008)433]. European Commission, Brussels.

CEC, 2011. White paper roadmap to a single European transport area – towards a competitive and resource efficient transport system (COM 2011/0144). European Commission, Brussels.

Chanaron, J.-J., Teske, J., 2007. Hybrid vehicles: a temporary step. International Journal of Automotive Technology and Management 7, 268–288.

Chau, K.T., Wong, Y.S., 2002. Overview of power management in hybrid electric vehicles. Energy Conversion and Management 43, 1953–1968.

Dijk, M., Kemp, R., 2012. Evolutionary policy options for steering the transition to low-carbon cars in Europe. In: Nilsson, M., Hillman, K., Rickne, A., Magnusson, T. (Eds.), Paving the Road to Sustainable Transport: Governance and Innovation in Low-Carbon Vehicles. Routledge, London, pp. 40–56.

Dyer, J.H., 1996. Does governance matter? Keiretsu alliances and asset specificity as sources of Japanese competitive advantage. Organization Science 7, 649–666.

Edquist, C., Hommen, L., Tsipouri, L.J., 2000. Public Technology Procurement and Innovation. Kluwer Academic, Boston.

Ehsani, M., Gao, Y., Gay, S.E., Emadi, A., 2005. Modern Electric, Hybrid Electric, and Fuel Cell Vehicles: Fundamentals, Theory and Design. CRC Press, Boca Raton, FL.

Energimyndigheten, 2011. Energy Indicators 2011. Energimyndigheten, Eskilstuna.

Etzkowitz, H., 2003. Beyond Humboldt: the entrepreneurial university, the third mission and the triple helix. VEST Journal of Science and Technology Studies 16, 21–45.

European Parliament and Council of the European Union, 2009. Directive 2009/28/EC of the European Parliament and of the Council of 23 April 2009 on the promotion of the use of energy from renewable sources and amending and subsequently repealing Directives 2001/77/EC and 2003/30/EC. European Commission, Brussels.

Geels, F., 2004. From sectoral systems of innovation to socio-technical systems. Insights about dynamics and change from sociology and institutional theory. Research Policy 33, 897–920.

Hillman, K., 2008. Environmental Assessment and Strategic Technology Choice – The Case of Renewable Transport Fuels. Chalmers University of Technology, Göteborg, Sweden.

Hillman, K., 2011. Governance of Innovation for Sustainable Transport: Biofuels in Sweden 1990–2010. Stockholm Environment Institute, Stockholm.

Hillman, K., Nilsson, M., Magnusson, T., Rickne, A., 2011. Fostering sustainable technologies – a framework for analysing the governance of innovation systems. Science and Public Policy 38.

Hooghe, L., Marks, G., 2003. Unravelling the Central State, but How? Types of Multi-level Governance. American Political Science Review 97, 233–243.
Howlett, M., 1991. Policy instruments, policy styles, and policy implementation: national approaches to theories of instrument choice. Policy Studies Journal 19, 1–21.
IEA, 2010. Energy Technology Perspectives 2010. Scenarios and Strategies to 2050. International Energy Agency, Paris.
IEA, 2011. Deploying Renewables: Best and Future Policy Practice. OECD/IEA, Paris.
Jacobsson, S., Bergek, A., 2004. Transforming the energy sector: the evolution of technological systems in renewable energy technology. Industrial and Corporate Change 13, 815.
Jacobsson, S., Lauber, V., 2006a. The politics and policy of energy system transformation – explaining the German diffusion of renewable energy technology. Energy Policy 34, 256–276.
Jacobsson, S., Lauber, V., 2006b. The politics and policy of energy system transformation – explaining the German diffusion of renewable energy technology. Energy Policy 34.
Jordan, A., Schout, A., 2006. The Coordination of the European Union. Oxford University Press, Oxford.
Kaiser, R., Prange, H., 2004. Managing diversity in a system of multi-level governance: the open method of co-ordination in innovation policy. Journal of European Public Policy 11, 249–266.
Kemp, R., Schot, J., Hoogma, R., 1998. Regime shifts to sustainability through processes of niche formation: the approach of strategic niche management. Technology Analysis and Strategic Management 10, 175–195.
Knill, C., Lenschow, A., 2005. Compliance, communication and competition: patterns of EU environmental policy making and their impact on policy convergence. European Environment 15, 114–128.
Kohler-Koch, B., Eisling, R. (Eds.), 1999. The Transformation of Governance in the European Union. Routledge, London.
Loorbach, D., 2010. Transition management for sustainable development: a prescriptive, complexity-based governance framework. Governance 23, 161–183.
Magnusson, T., 2011. Governance of Innovation for Sustainable Transport: Hybrid-Electric Vehicle Technology in Sweden 1990–2010. Stockholm Environment Institute, Stockholm.
Magnusson, T., Berggren, C., 2011. Entering an Era of Ferment – radical vs. incrementalist strategies in automotive powertrain development. Technology Analysis and Strategic Management 23, 313–330.
March, J., Olsen, J., 1989. Rediscovering Institutions: The Organizational Basis of Politics. The Free Press, New York.
Markard, J., Truffer, B., 2008. Technological innovation systems and the multi-level perspective: towards an integrated framework. Research Policy 37, 596–615.
Midttun, A., Gautesen, K., 2007. Feed in or certificates, competition or complementarity? Combining a static efficiency and a dynamic innovation perspective on the greening of the energy industry. Energy Policy 35, 1419–1422.
Mikler, J., 2005. Institutional reasons for the effect of environmental regulations: passenger car CO_2 emissions in the European Union, United States and Japan. Global Society 19, 409–444.
Newell, P., 2008. The political economy of global environmental governance. Review of International Studies 34, 507–529.
Newell, P., Jenner, N., Baker, L., 2008. Governing Clean Development: A Framework for Analysis. University of East Anglia, Norwich.
Nilsson, M., Nilsson, L.J., Stripple, J., Hildingsson, R., 2011. The missing link: bringing institutions and politics into energy future studies. Futures 43, 1117–1128.
Nygaard, S., 2008. Co-evolution of Technology, Markets and Institutions: the Case of Fuel Cells and Hydrogen Technology in Europe. Centre for Innovation, Research and Competence in the Learning Economy (CIRCLE), Lund University, Lund, p. 287s.
Peters, G. (Ed.), 2000. Institutional Theory in Political Sciences: The New Institutionalism. Continuum, New York.
Peters, G.B., 1998. Managing horizontal government: the politics of coordination. Public Administration 76, 295–311.
Pierre, J., Peters, G., 2006. Governing Complex Societies. Palgrave Macmillan, Basingstoke.
Scharpf, F., 1993. Coordination in hierarchies and networks. In: Scharpf, F. (Ed.), Games in Hierarchies and Networks. Campus, Frankfurt, pp. 125–165.
Scott, W.R., 1995. Institutions and Organizations. Sage, Thousand Oaks, California, US.
Steinmo, S., Thelen, K., Longstreth, F. (Eds.), 1992. Structuring Politics: Historical Institutionalism in Comparative Analysis. Cambridge University Press, Cambridge.
Treib, O., Bähr, H., Falkner, G., 2007. Modes of Governance: towards conceptual clarification. Journal of European Public Policy 14, 1–20.